

Influence of Large Converter Based Loads on Oscillations in Transmission Grids

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Abstract—The power system of today's world is constantly changing. With the shift towards electricity generation from renewable sources such as wind- and sun-power, power-electronics based converters have more and more impact on the power grid. But not only the generation side of the power system is changing, also loads are evolving. On the consumption side, there is a rapid increase in loads that are connected to the grid using power electronic converters, some of them can reach a very high power. Converter-based loads have different characteristics than traditional loads, leading to new challenges when connecting them to the power grid. One big problem faced by grid operators and planners around the world, is the ability of large, converter-based loads to influence oscillation phenomena in different frequency ranges. This report will review what kind of converter-based loads there are in the transmission grid, their characteristics, how they can influence oscillations and strategies to assess and manage the risks coming from oscillations caused by converter-based large loads.

Index Terms—converter-based large loads, power system oscillations, transmission grid dynamics

I. INTRODUCTION

The energy transition towards renewable generation and the growing electrification of demand has a great impact on the behaviour and dynamics of our power grid. Not only generation sources use power electronic converters to interface with the grid, converter-based loads (CBLs) are also emerging at a very fast rate. Connecting these loads to the transmission system poses several challenges and difficulties because of the inherently different behaviour of converter based loads compared to other large loads that have been studied for a long time and are generally well understood. Currently there is a lot of research in the direction of CBLs, with some organizations forming large load task forces, for example the North American Electric Reliability Corporation (NERC)[1], specifically focusing on the impact of CBLs on the power grid. Connecting CBLs to the grid introduces several risks to the grid. Those include aspects like frequency-, voltage- and rotor angle-stability or harmonics [1]. Another risk that is drawing the attention of researchers and grid operators around the world is the ability of CBLs to influence oscillation phenomena in the power grid. Oscillations in power systems have been observed and studied for a long time. Uncontrolled oscillations can generally lead to unstable flows of energy, which can potentially damage equipment, trip protection devices and lead to cascading outages[2]. The ability of CBLs to

influence oscillation phenomena already showed in a number of recent incidents.

One example is the 14,7 Hz oscillation event in the Dominion Energy System of the North American power grid. The oscillation in the system voltage was observed after the ramp-down of four hydro-generation units in a region with a high concentration of large, converter-based loads, namely data centers. The oscillation lasted for a duration of 2 hours before it could be damped by adjusting the setpoint of a nearby STATCOM. After an investigation, the origin of the oscillation was identified to be the uninterrupted power supply of a data center.[3]

In 2024, the Electric Reliability Council of Texas noticed power oscillations of a CBL with a magnitude of 50 MW and a frequency of 23 Hz, after the load exceeded a certain threshold in power consumption. It was determined that this poses a reliability risk to the grid because load oscillations in this frequency range could potentially damage nearby synchronous generators and the load had to reduce its consumption to a level where no oscillations were noticeable. In this case, the cause of the oscillations was determined to be old software on equipment in the load.[2]

These recent incidents show that the ability of CBLs to influence oscillation phenomena in different frequency ranges poses real risks to the power grid. To better understand the causes and risks coming from CBL-influenced oscillation phenomena, this report will answer the following five questions:

- 1) What kinds of converter-based large loads are becoming relevant in transmission grids?
- 2) How do converter-based large loads behave differently from classical loads during disturbances?
- 3) Under which grid and operating conditions can these loads contribute to oscillation problems?
- 4) Which oscillation phenomena are most likely to be influenced by converter-based large loads?
- 5) How should grid planners and operators assess and manage oscillation risks associated with converter-based large loads?

II. KINDS OF CONVERTER-BASED LARGE LOADS IN THE TRANSMISSION GRID

A. Industries with a high amount of variable frequency drives

A variable frequency drive (VFD) is a power electronic device that converts AC power from the grid to AC power with a variable frequency to drive electric machines with a variable speed. In today's automated industry, VFDs are used for many different applications. In industries like mining and water treatment, VFDs drive large induction motors that power for example conveyor belts. The motors can reach a power of several megawatts and are therefore often fed by medium voltage converters. A plant that uses a lot of these medium voltage drives can easily reach a total load of several tens of megawatts and therefore has a significant impact on the power grid.

B. Electrified railway

Electrified passenger- and freight-trains are an environmentally friendly way of transporting people and goods. In Europe, 60 % of the rail network is already electrified, with 80 % of rail traffic running on these lines[4]. A converter in an electric train converts the AC power from the overhead line above the train track and converts it to AC with variable frequency, similar to the VFDs used in industry. Modern electric trains, for example the German high-speed train ICE can reach a power of up to 11,5 MW[5].

C. Electric vehicle charging hubs

The electrification of cars and other road vehicles is growing at a very high speed. This comes with the need for a reliable and fast charging infrastructure. In an EV-charging hub up to hundreds of high power EV chargers are concentrated in one place. DC charging is typically used for the fast charging of electric vehicles. This means that the AC power from the grid is converted to DC power in the charging station and gets then transferred to battery of the vehicle. The power of a DC-fast-charging station typically reaches up to 350 kW, with charging power increasing in the future. The biggest EV charging hub in the world consists of 258 fast chargers with a combined load of 93 MW[6].

D. Hydrogen electrolyzers

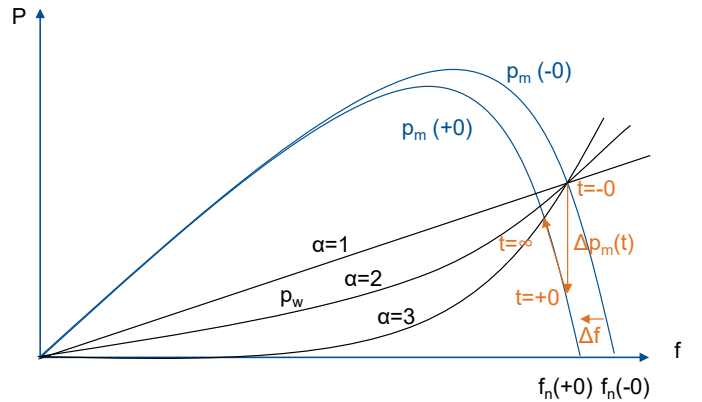
Green hydrogen plays an important role in the decarbonization of energy intensive industries like the chemical or steel industry and therefore has the potential to help the transition into a climate-neutral future. Green hydrogen is produced by water electrolysis, using renewable energy sources. The German grid development plan forecasts the load of hydrogen electrolyzers connected to the grid to be between 31 GW and 70 GW by the year 2045[7]. In an electrolysis plant, AC power from the grid is converted into DC power that can be used for water electrolysis using mostly thyristor-based rectifiers. In the future, electrolysis plants could reach a load of up to several hundreds of megawatts.

E. Data centers

With the growing digitalisation of our world, there is an increasing need for digital infrastructure and computing power in the form of data centers. The emergence of artificial intelligence further amplifies this demand, while being even more energy intensive than traditional data centers. Germany is already the largest location for data centers in Europe with 2000 data centers and a combined load of 2,7 GW in 2024. This load is expected to increase to over 10 GW by the year of 2045[8]. The power of the largest AI-data centers already reaches more than 1 GW, with facilities with a load of multiple gigawatts being planned and constructed in the future. Converter based loads of this scale have not yet been seen before and pose significant challenges when connecting them to our power grid.

III. BEHAVIOUR OF CONVERTER-BASED LOADS COMPARED TO CLASSICAL LOADS

The difficulties coming with the connection of CBLs to the power grid come from their naturally different behaviour compared to conventional loads. Conventional loads like induction motors have a stabilizing effect when a voltage or frequency disturbance occurs in the grid. This means that they reduce their active power draw if there is a drop in voltage or frequency. This property is called the self regulating effect and it has a stabilizing effect on the disturbance in the grid.[9] Induction motors also provide mechanical inertia due to the kinetic energy stored in the machines rotating mass, helping to limit the rate of change of frequency (RoCoF) during a disturbance. In Figure 1, the frequency dependence of an induction motor is shown.



Working machines	α
Translatory motion	1
Laminar flows	2
Turbulent flows	3

Fig. 1: Frequency dependence of induction motor [9]

Other conventional loads like resistive loads or loads that draw a constant current can generally be described with a load model such as the ZIP-model. The ZIP model describes the dependence of the load on the voltage. It consists of a constant

impedance part with a quadratic dependence on voltage, a constant current part with a linear dependence and a constant power part independent to the voltage of the system.[10]

$$P = P_0 \cdot \left[Z_p \cdot \left(\frac{U}{U_N} \right)^2 + I_p \cdot \left(\frac{U}{U_N} \right)^1 + P_p \cdot \left(\frac{U}{U_N} \right) \right] \quad (1)$$

While most conventional loads can be described by such a load model, CBLs have characteristics that make representing them by a conventional load model difficult or inaccurate. One reason for this is the constant power characteristic of CBLs. In the event of a disturbance such as a voltage sag, this means that the load increases its current draw to compensate for the lower voltage. This results in the voltage sag being amplified by the load now drawing more current. This negative incremental impedance of CBLs has a destabilizing effect on the grid because neagative damping is introduced to the system across all frequency ranges.[11] Figure 2 shows the I-V-curve of a constant power load and the definition of the incremental impedance. Many CBLs like data centers or hydrogen electrolyzers are very sensitive to voltage deviations and may disconnect if the system voltage is too high or low. To avoid dangerous over-voltage and over-frequency caused by the diconnection of large loads, grid operators put fault-ride-through requirements on large load facilities[12].

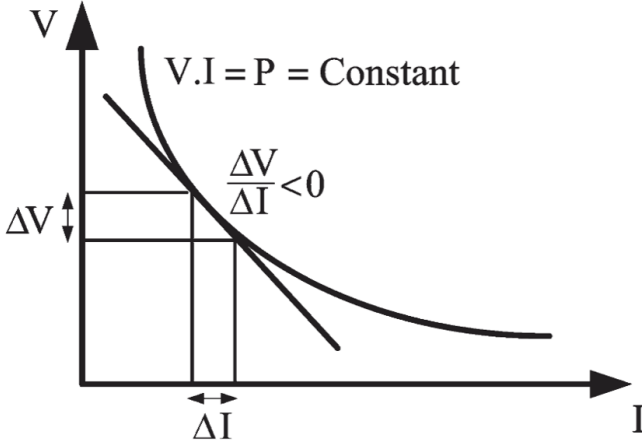


Fig. 2: I-V curve of a constant power load [13]

CBLs also do not provide any inertia to the grid to limit frequency excursions. An induction motor fed by a variable frequency drive has a rotating mass just like an induction motor that is directly connected to the grid, but there is no electromechanical coupling between rotation of the machine and the grid frequency, as they are decoupled through the DC-link of the converter.

Another characteristic that makes describing CBLs by static load models difficult are the fast dynamic interactions of power electronic converters. Modern semiconductor switches used in converters operate with a high switching frequency, introducing high frequency harmonics into the power grid. The converters are controlled by multiple fast control loops

with different bandwidths. This along with the naturally non-linear behaviour of power electronic converters makes them very difficult to compare to the conventional loads and load models.

IV. GRID AND OPERATING CONDITIONS LINKED TO OSCILLATION PROBLEMS

To understand how CBLs can influence oscillation phenomena, it is important to understand what kind of grid and operating conditions are benficial for the forming of oscillations.

A. Weak grid conditions

The strenght of a power grid is generally described by its short circuit ratio (SCR). The simplest definition of the SCR is the ratio of the short circuit apparent power to the power of a load at a certain point of interconnection:

$$SCR = \frac{S_{SC}}{P_{Load}} \quad (2)$$

A low short circuit ratio corresponds to a low short circuit current in the system. The short circuit current is mostly influenced by the grid impedance and the generators in the system. The impedance of the grid increases when it is weakly interconnected or features very long transmission lines reaching into remote areas. Synchronous generators also play an important part contributing to the short circuit ratio, as they can deliver short circuit currents multiple times higher than their rated current. Converter-based generators provide no substantial short circuit current, because the short circuit current is restricted by device limitations of semmiconductor switches. This leads to weak grid conditions in regions with a high penetration of power electronic converters. [14]

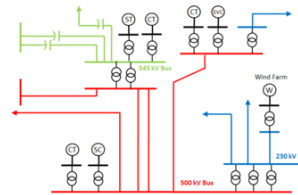


Fig. 3: Example of a strong grid

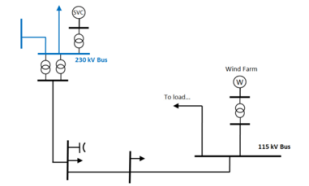


Fig. 4: Example of a weak grid

Weak grid conditions, which are described by a SCR lower than 3, generally indicate a high sensitivity of the voltage to changes in active and reactive power changes. This leads to reduced voltage stability of the system.[14] There is also an increasing risk of oscillations arising from interaction between controllers of different converters[15]. Like this, CBLs can contribute to oscillation problems in areas with weak grid conditions. If large loads are also geographically concentrated, for example electrolyzer facilities in northern germany, it can amplify the described problem.

B. High penetration of power electronic converters

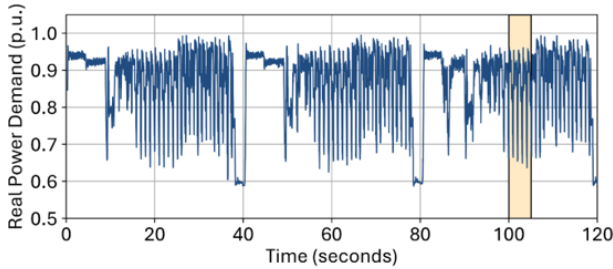
A high penetration of power electronic converters in a grid not only influences the SCR. A lack of synchronous generators in a system leads to a lack of mechanical inertia, which is important for frequency stability. With low mechanical inertia available, frequency excursions get larger and the RoCoF is faster, potentially leading to sympathetic tripping of other inverter-based-resources (IBR) and cascading grid outages. This happens when the RoCoF causes the phase-locked-loop, which grid-following inverters use to track the phase angle of the voltage, fails. Additionally, control loop interactions between different converters can cause oscillatory behaviour.[15]

C. High loading conditions of the grid

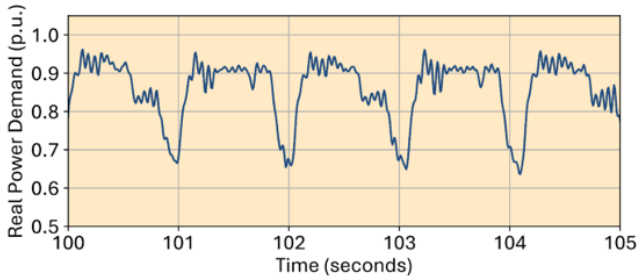
When a power grid experiences high loading conditions, it operates more closely to its stability margins. Under high loading, the angle difference in a synchronous generator is increased. Under those conditions the synchronizing torque in the generator is lower, which leads to reduced damping and a higher risk of oscillations occurring.[16]

D. Load fluctuations in AI data centers

Data centers for artificial intelligence applications experience periodic fluctuations in active power consumption. In parallel computing tasks, such as the training of AI-models, large amounts of graphical processing units get activated and deactivated in a synchronous way. The swings in power demand may be as high as from 10 % to 100 % of the load in a very short time frame.[17]



(a) Load profile example over 2 minute period



(b) Load profile example over 5 second period

Fig. 5: Example of an AI training load profile [1]

As it can be seen in figure 5, the load variations seen in AI training can be bi-periodic[17]. There are variations in power with lower frequency and a high amplitude and variations with

a higher frequency but a lower amplitude. These so-called forced oscillations can excite existing oscillation modes in a system and cause oscillation problems[1].

V. OSCILLATION PHENOMENA INFLUENCED BY CONVERTER-BASED LARGE LOADS

A. Electromechanical Oscillations

Electromechanical oscillations are a widely known phenomena in electrical power systems. The frequency of these oscillations typically lies in the range of 0,1 Hz to 2 Hz. If a synchronous generator rotating with a constant speed is subjected to a small disturbance, it experiences a synchronizing torque that accelerates or decelerates the rotating mass of the generator. During this acceleration and deceleration active power is exchanged between the generator and other generators in the system in an oscillating way. A system of multiple generators and transmission lines can be described with a spring-mass system analogy: the generators are represented with masses and the transmission lines by the springs. If a mass now experiences a disturbance like an external force, the system starts oscillating. The frequencies at which the system is oscillating are called modes. Modes between closely coupled generators in smaller areas are called local modes, modes between generators that are geographically far apart and coupled with long transmission lines are called global modes. If the modes are not damped sufficiently, dangerous power oscillations can happen that lead to overloaded transmission lines or the loss of synchronism in generators.[16]

Large loads with periodic variations in power consumption can excite the local- or inter-area-modes of a system. As discussed in chapter 4 D, data centers can have bi-periodic variations in their power consumption. The lower frequency variations are in a frequency range relevant for electromechanical phenomena and could lead to excitation of oscillation modes.

B. Subsynchronous oscillations

The rotor of a turbine-generator system consists of several sections with different masses. A system like this has multiple torsional modes that can be excited by electrical disturbances at the output of the generator. If those modes get excited, torsional oscillations result between the different sections of the rotor, creating a torsional force on the rotor shaft.[18] The forces produced on the rotor shaft can lead to material fatigue or failure. This phenomenon is called subsynchronous torsional interaction (SSTI). The high frequency load variations in data converters discussed in section 4 D lie in a frequency range relevant for SSTI, potentially producing torque oscillations dangerous for the shaft of generators that are located close to the large load[12].

Another oscillation phenomena in the subsynchronous frequency range is the subsynchronous control interaction (SSCI). The control system of a large load could interact with the controllers of other nearby converters, such as HVDC converters or renewable generation. SSCI involve also interactions of

converters with series compensated transmission lines, creating points of resonance.[12] [19]

C. Voltage oscillations

As it was discussed in chapter 4 A, weak grid conditions lead to a high sensitivity of the voltage to changes in active power. Like this, load variations in data centers could cause fluctuations in the voltage. This may cause noticeable flickering of nearby electric lighting.[20]

D. Wideband oscillations

Wideband oscillations include every oscillation phenomena up to a frequency range of up to several kHz, thus including oscillations caused by converter harmonics. Wideband oscillation phenomena are mostly observed and studied in power systems with a high penetration of renewable generation.[21] There is no evidence of wideband oscillations linked to CBLs yet, but being converter-based systems, they also inject harmonics into the grid that could lead to resonances and oscillations in higher frequencies.

VI. ASSESSING AND MANAGING RISKS OF OSCILLATIONS

It has become clear that oscillation phenomena influenced by converter-based large loads can be a big risk to reliability and stability of the power system. Grid operators will have to assess and manage the risks coming from large load induced oscillations to safely connect them to the power grid. Dynamic simulations and general requirements for CBLs can help to limit the risks coming from oscillations and will be discussed in the following chapter.

A. Dynamic simulations

A reliable operation of the electrical power system requires simulating its behaviour during the planning phase to identify certain risks and find strategies to mitigate these risks. With the emergence of large CBLs that feature less predictable dynamic responses than conventional loads, there is a need for detailed dynamic simulation models that can accurately represent their behaviour. Through performing dynamic simulations, a variety of stability phenomena like rotor angle stability, voltage stability, frequency stability and oscillations can be studied to examine the impact of a large load on the grid. For dynamic simulations, RMS and electromagnetic transient (EMT) models are used. To this day, as the research activity about large loads is still in its early stages, there are no standardized library models implemented in the simulation tools used in the industry. This means that custom site-specific simulation models have to be developed.[22]

RMS simulations are based on a positive sequence phasor and are used to study lower frequency dynamics up to 10 Hz in the grid. Effects that can be studied with RMS simulations include electromechanical oscillations or load transients. RMS models of large loads have to represent the actual composition of the load. This includes the power electronic converters, but also motors and static loads that are present in the load. It is also necessary to represent any control loop of the load that

falls into the frequency bandwidth of under 10 Hz, for example active power control or power factor control. The faster inner control loops can typically be omitted in RMS models. RMS models are limited in their bandwidth, so higher frequency dynamics like subsynchronous oscillations can not be studied. The effect of unbalanced voltage or frequency also can not be represented with RMS models.[22]

The EMT model is based on solving the differential equations of a power system in the time domain. This makes it the most detailed model, but also the most computationally intensive. With the EMT model, higher frequency dynamics like subsynchronous oscillations and control interactions can be studied. For large load modeling, it is important that the EMT model represents every significant component of the load. This includes for example converters, cooling systems, motor drives, uninterrupted power supplies and backup generators. Unlike in the RMS model, every control system and protection device should be part of the model. Control and protection parameters like controller gains should be exactly like in the real load. If the load experiences periodic variations in its power consumption, a realistic load profile with a high resolution should be used.[22]

Dynamic simulations with RMS- and EMT-models can also be combined with an impedance-based analysis for the investigation of oscillation phenomena. A passive frequency sweep models the network using R, L, and C components, a dynamic frequency scan also includes controller dynamics. A passive and dynamic frequency can be used to identify frequency regions of resonance and negative damping in a network. Afterwards, an EMT simulation can be performed to see if oscillations form at those regions and strategies for mitigation can be found.[23]

B. General requirements for large, converter-based loads

To ensure a safe operation of large CBLs in the grid, grid operators and planners also have to put requirements on large load facilities. Those requirements include aspects like fault-ride-through behaviour, active power recovery and reactive power requirements. Concerning oscillations, the load ramp rates and cycling is most important. The high and load frequency load cycling present in data centers should be limited in a way so it does not excite any known electromechanical or subsynchronous oscillation modes. Many grid operators already stated limits for load ramp rates. One example is the American Transmission Company ATC, which requires that active power variations in large loads must remain under 25 MW if they happen in a time period of less than 5 seconds. Large load facilities could also be required to install suitable compensation devices on-site, like E-STATCOMs or battery storage systems.[12]

For the monitoring of oscillation phenomena, the installation of measurement devices like phasor measurement units (PMU) and point-on-wave measurements (POW) at large load facilities can be required. This enables grid operators to accurately monitor load induced oscillations in various frequency ranges for reliable operation.[12][19]

VII. CONCLUSION

In the future, the number of large, converter-based loads connected to the power grid will grow rapidly, and with that the amount of disturbance events linked to them. CBLs feature distinct characteristics that are not comparable to conventional loads and can influence various oscillation phenomena that pose a big risk to the reliable and stable operation of the power grid. The risk of oscillations is especially high in weak and converter dominated grids and most prominent in AI-data centers with periodic variations in their load profiles. Grid operators can assess and manage the risks of oscillations before the connection to the grid by performing dynamic simulations and putting requirements on large load facilities to ensure that they are operating in a safe way.

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